

7

CLOCKS

7.1. Introduction

In a clock the most important hands are the minutes hand and the hours hand. Whatever may be the shape of the dial they move in a circular track.

The total angle of 360 degrees in a watch is divided into 12 sectors, one for each hour.

So one hour sector = $\frac{360}{12} = 30$ degrees.

For every one hour (60 min),

- The minutes hand moves through 360 deg.
- The hours hand moves through 30 deg.

So for every minute,

- The minutes hand moves through 6 deg
- The hours hand moves through 0.5 deg.

They move in same direction. So their relative displacement for every minute is 5.5 deg.

This 5.5 deg movement constitutes the movements of both the hands.

So for every minute both the hands give a displacement of 5.5 deg.

Note:

1. Between every two hours i.e., between 1 and 2, 2 and 3 and so on the hands of the clock coincide with each other for one time except between 11, 12 and 12, 1.
In a day they coincide for 22 times.
2. Between every two hours they are perpendicular to each other two times except between 2, 3 and 3, 4 and 8, 9 and 9, 10.
In a day they will be perpendicular for 44 times.
3. Between every two hours they will be opposite to each other one time except between 5, 6 and 6, 7.
In a day they will be opposite for 22 times.

Examples: At what time between 5 and 6 will the hands of the clock coincide?

Solution: At 5 the angle between the hands is 150 deg.

To coincide, they collectively have to travel this distance. Every minute they travel 5.5 deg.



So no. of minutes required to coincide = $\frac{150}{5.5} = \frac{300}{11} = 27\frac{3}{11} \text{ min.}$

Examples: At what time between 6 and 7 will the hands be perpendicular?

Solution: At 6 the angle between the hands is 180 deg.

To form 90 deg they have to cover 90 deg (out of 180 if 90 is covered 90 will remain)

So no. of minutes required = $\frac{90}{5.5} = \frac{180}{11} = \frac{164}{11} \text{ min.}$

But they will be perpendicular for two times. The second one will happen after the minutes hand crosses the hours hand and then for 90 deg.

So it has to travel $180 + 90 = 270$ deg.

So time = $\frac{270}{5.5} = \frac{540}{11} = 49\frac{1}{11} \text{ min.}$

Examples: What is the angle between the hands of the clock at 3.45?

Solution: At 3, the angle between the hands = $A = 90$ deg.

In 45 min the hands will move angle of $B = 45 \times 5.5$ deg (since 5.5 deg for 1 min)

$B = 247.5$ deg.

Required angle = $A \sim B = 157.5$ deg.

Examples: What is the angle between the hands at 4.40?

Solution: At 4 the angle between the hands, $A = 120$ deg.

In 40 min, $B = 40 \times 5.5 = 220$ deg.

The required angle = $A \sim B = 100$ deg.

Examples: A clock loses 5 min for every hour and another gains 5 min for every hour. If they are set correct at 10 am on Monday then when will they be 12 hrs apart?

Solution: For every hour watch A loses 5 min and watch B gains 5 min.

So for every hour they will differ by 10 min.

For 12 hrs (720 min) difference between them the time required = $\frac{720}{10} = 72 \text{ hrs}$

So they will be 12 hrs apart after 3 days i.e., at 10 am on Thursday.

